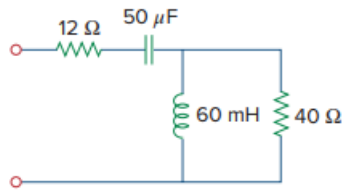


Problem

At $\omega = 377$ rad/s, find the input impedance of the circuit shown in Fig. 9.63.

Figure 9.63:



Step-by-step solution

Step 1 of 4

The following is the given circuit diagram:

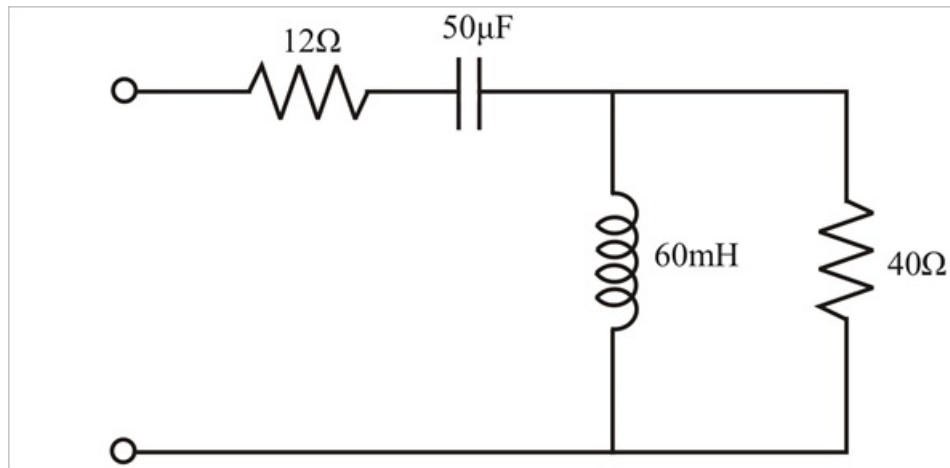


Figure 1

Step 2 of 4

It is given that,

The angular frequency is, $\omega = 377$ rad/sec

The impedance of the 60 mH inductor is,

$$\begin{aligned} jX_L &= j\omega L \\ &= j(377)(60 \times 10^{-3}) \\ &= j22.62 \, \Omega \end{aligned}$$

The impedance of the 50 μ F capacitor is,

$$\begin{aligned} -jX_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(377)(50 \times 10^{-6})} \\ &= -j53.05 \, \Omega \end{aligned}$$

Step 3 of 4

The following is the modified circuit of Figure 1 in frequency domain.

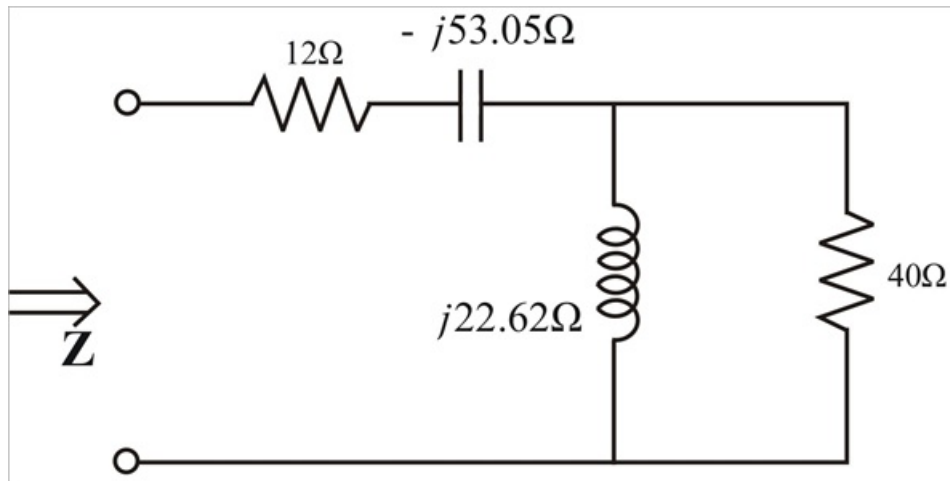


Figure 2

Step 4 of 4

From the Figure 2, the equivalent impedance is,

$$\begin{aligned} \mathbf{Z} &= 12 - j53.05 + (40 \parallel j22.62) \\ &= 12 - j53.05 + \frac{(40)(j22.62)}{40 + j22.62} \\ &= 12 - j53.05 + \frac{j904.8}{40 + j22.62} \\ &= 12 - j53.05 + 9.69 + j17.14 \\ &= 21.69 - j35.91 \, \Omega \end{aligned}$$

Therefore, the input impedance of the circuit, $\mathbf{Z}$ is $\boxed{21.69 - j35.91 \, \Omega}$.